MARIO ALEJANDRO MARROQUÍN

El Salvador · mario.marroquin.2007@gmail.com · linkedin.com/in/mario-alejandro-marroquín-305a843a0
github.com/mariomarroquin3 · orcid.org/0009-0000-7244-1997 · Portfolio: mariomarroquin3.github.io

Machine learning practitioner (PyTorch, scikit-learn) with original research in tail-risk forecasting and medical-image classification, and a national AI hackathon win. I make the design decisions and build with AI agents.

EDUCATION

Complejo Educativo "Thomas Jefferson" El Salvador
Technical Baccalaureate in Technology Infrastructure and IT Services *Expected 2026*
- Straight-10 grades and first place in class every year.

EXPERIENCE

Accounting Assistant, Process Automation Feb 2026 – Present
El Salvador
- Built Partidas DTE, a desktop app that reads identifiers from accounting entries (PDF) and merges the matching electronic invoices into one file, cutting a task that took a full afternoon (~4 hours) to under 1 minute (over 99% faster), recurring monthly across multiple clients.

RESEARCH & PROJECTS

CryptoRisk github.com/mariomarroquin3/cryptorisk
- Compared 20+ volatility and tail-risk models (GARCH family, HAR, CAViaR, MS-GARCH, RF-QR, LSTM) for 1-day VaR and Expected Shortfall on BTC and ETH with a 500-day walk-forward protocol; evaluated with Kupiec, Christoffersen, Dynamic Quantile, Acerbi-Szekely, FZO, Diebold-Mariano and MCS tests.

Grainy · Coffee quality prediction National AI Hackathon winner (2025)
- Built a Streamlit app that predicts coffee quality with Random Forest regressor and classifier models, generates PDF reports and includes a Gemini chatbot (RAG with FAISS) over a technical guide.

RetinoVisionAI EUREKA challenge
- Trained an EfficientNet-B3 classifier for diabetic retinopathy with augmentation, illumination normalization and stratified cross-validation; added Grad-CAM interpretability and a web app (PyTorch, Flask).

Financial data synthesizer (VAE) github.com/mariomarroquin3/data_generator
- Built a Variational Autoencoder (PyTorch) that generates synthetic financial transactions for testing accounting software without exposing real data, with a Streamlit app and PCA validation (Code in Place final project).

Data Miner · multi-agent team (BoxLang AI Hackathon 2026) github.com/mariomarroquin3/dataMiner
- Team of four; I built the agent architecture and backend: coordinator plus three agents, personal-data anonymization, read-only SQL/MCP and re-execution of the SQL behind each figure; extracted the module github.com/mariomarroquin3/bx-datatools.

HONORS, LEADERSHIP & PROGRAMS

- Winner, first National Artificial Intelligence Hackathon (2025): coffee quality classification with a Random Forest model.
- Multiple wins in district and departmental public speaking contests.
- Mathematics olympiads, two consecutive years.
- Member, Cohere Labs Open Science Community and Google Developer Program; participant, ML Summer School 2026.
- In progress: GCI Global 2026 credit-risk competition (Matsuo Lab, University of Tokyo); Kaggle Enveda CASMI 2026; CUBO+ and CUBO_ai; Stanford Probability for AI.

CERTIFICATIONS

- AWS: ML Engineer Associate Curriculum Overview; Introduction to Machine Learning; SimuLearn Cloud Computing Fundamentals (2026).
- Platzi: Linear Algebra for ML; Math Functions for Data Science and AI; Math Language and Notation; Math Fundamentals (2026).
- Google Developer Program badges: ML Crash Course (Linear and Logistic Regression), Gemini Enterprise Agent Ready, Google Cloud & NVIDIA community member, Chrome DevTools, DOM Detective, Android Studio.
- Direct English, Intermediate (2024).

SKILLS & LANGUAGES

Programming: Python, SQL, LaTeX. Working knowledge with AI assistance: TypeScript, JavaScript/React, Java, VBA; learning R.
ML & Data: PyTorch, scikit-learn, statsmodels, arch, Pandas, NumPy, Matplotlib, Streamlit, Flask.
Languages: Spanish (native); English (intermediate certificate 2024, advanced level in preparation).